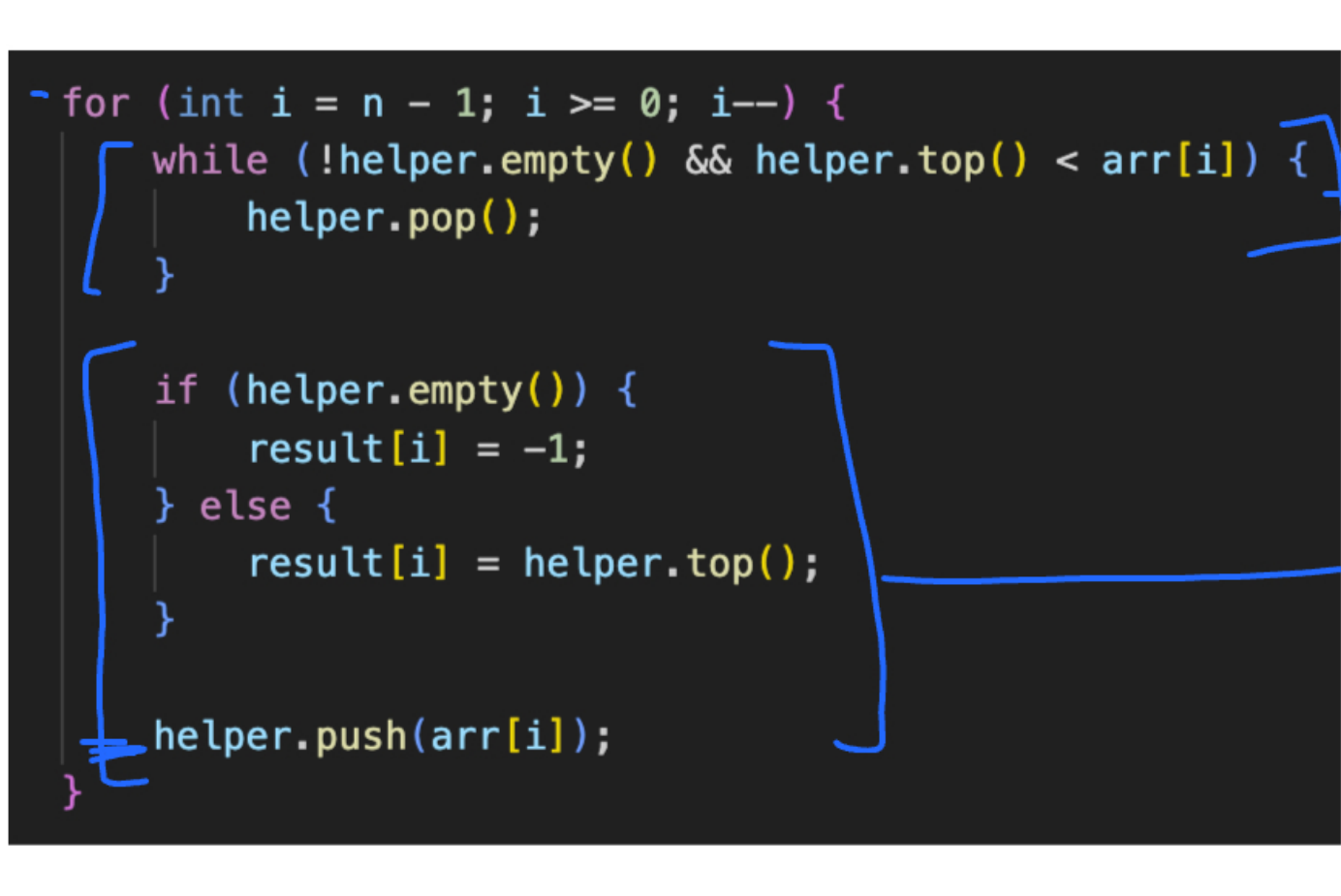


```
for (int i = n - 1; i >= 0; i--) {
    while (!helper.empty() && helper.top() < arr[i]) {
        helper.pop();
    }

    if (helper.empty()) {
        result[i] = -1;
    } else {
        result[i] = helper.top();
    }

    helper.push(arr[i]);
}
```

→ O(n)

→ O(n)

TC: O(n+n) = O(n).

6 0 1 7

[5 4 3 2 1]

-1 -1 -1 -1 -1

5 ~~4~~ ~~3~~ ~~2~~ ~~1~~

[6, 2, 5, 4, 5, 1, 6]

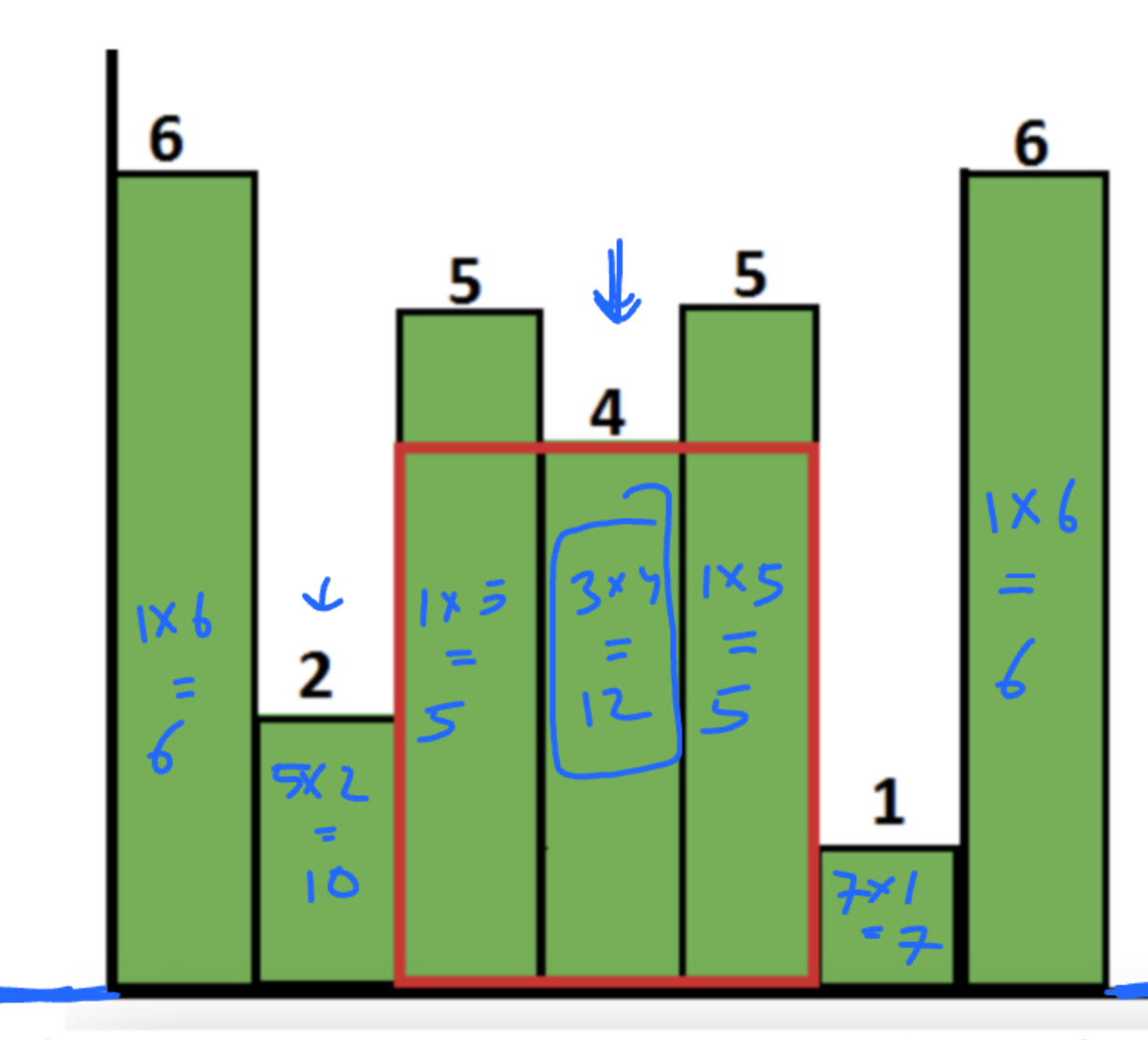
~~6~~ 8 ~~10~~ 12

i: 0 → n-1

j: i → n-1

TC: O(n²).

AS: O(1)



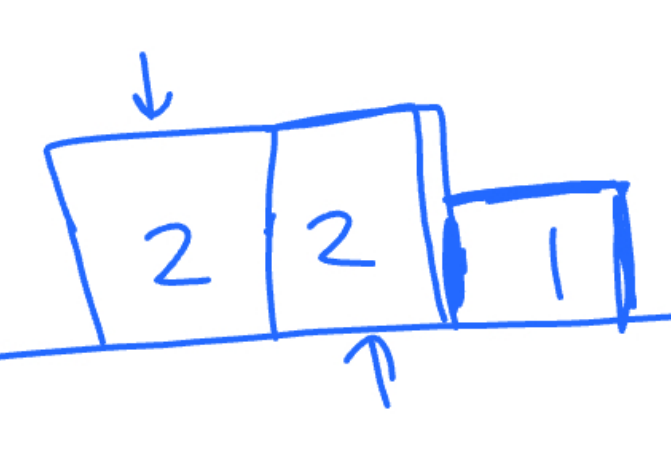
n=7

	0	1	2	3	4	5	6
arr:	6	2	5	4	5	1	6
prevSm:	-1	-1	1	1	3	-1	5
nextSm:	1	5	3	5	5	7	7
width:	1	5	1	3	1	7	1
Area:	6	10	5	12	5	7	6

[width = nextSm[i] - prevSm[i] - 1]

TC: O(n).

AS: O(n).



---

0 1 2 3 4 5 6

[10, 20, 30, 50, 10, 70, 30].

10 20 30 10 10 30

k=1 → 70

k=2 → 30

k=3 → 20

k=4 → 10

k=5 → 10

k=6 → 10

k=7 → 10

1 ≤ k ≤ n

BF. [k=1 → n [(n-k+1) * K]] TC = O(n²) AS = O(n).

1 → n

2 → n-1

3 → n-2

⋮

k → n-k+1

[10, 20, 30, 50, 10, 70, 30].

0	1	2	3	4	5	6
~~50~~	30	20	10	10	10	10
70	~~30~~					~~10~~

TC: O(n).

AS: O(n).

	0	1	2	3	4	5	6
arr	10	20	30	50	10	70	30
prev:	-1	0	1	2	-1	4	4
next:	7	4	4	4	7	6	7
width:	7	3	2	1	7	1	2

width = nextSm[i] - prevSm[i] - 1

0	1	2	3	4	5	6
~~50~~ ~~50~~ 70	~~30~~ 30	~~20~~ 20	~~10~~ 10	~~10~~ 10	~~10~~ 10	~~10~~ 10

Queue.

FIFO: First In First Out

```
class Queue {
  private:
      L.L.
  public:
      push()
    - pop()
      front()
      size()
      isEmpty()
}
```

Push(1)
push(2)
- front() — 1
pop()
pop()
push(3).
front() —— 3

front [~~1~~ | ~~2~~ | 3]

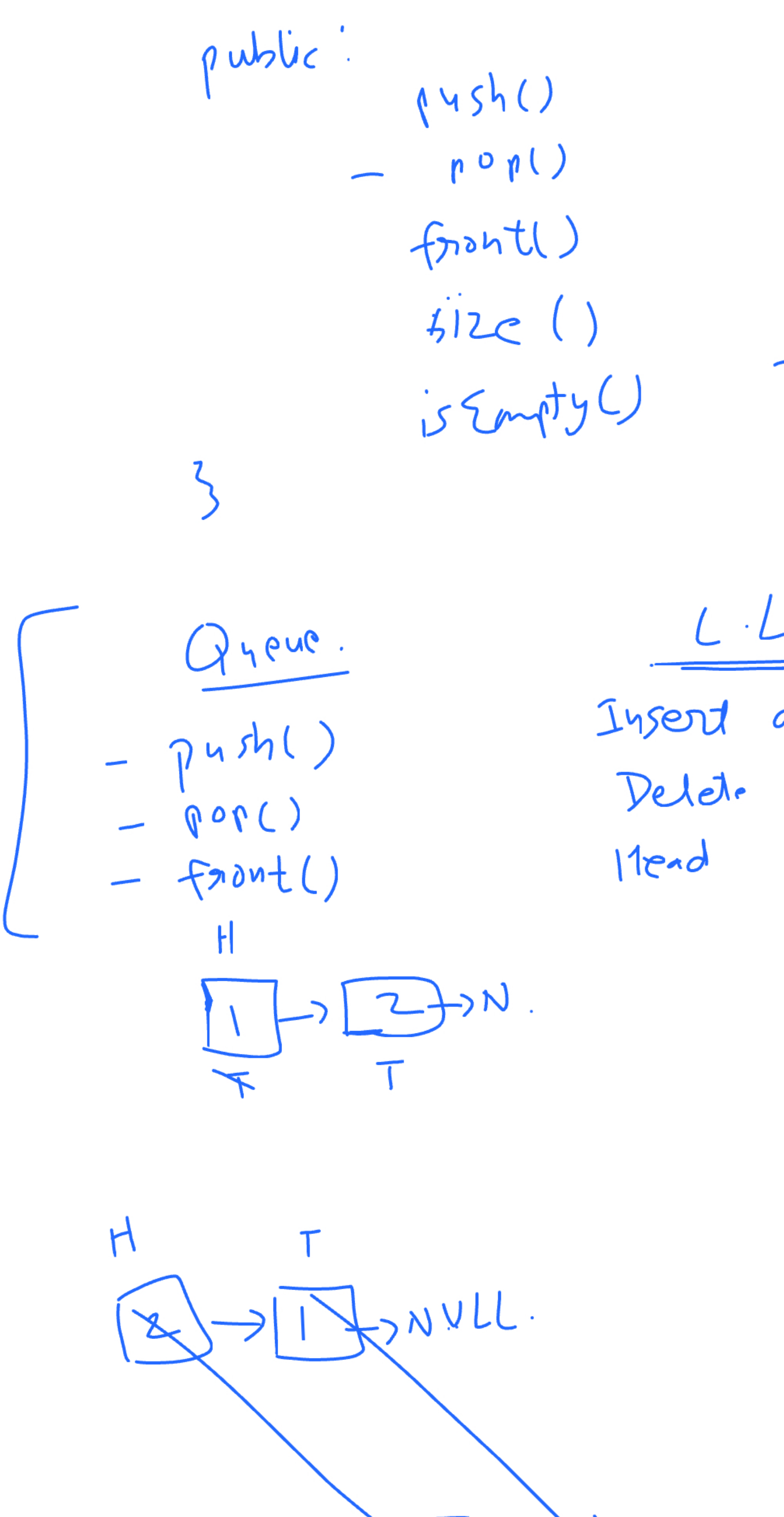
[Queue.
- push()
- pop()
- front()]

L.L.

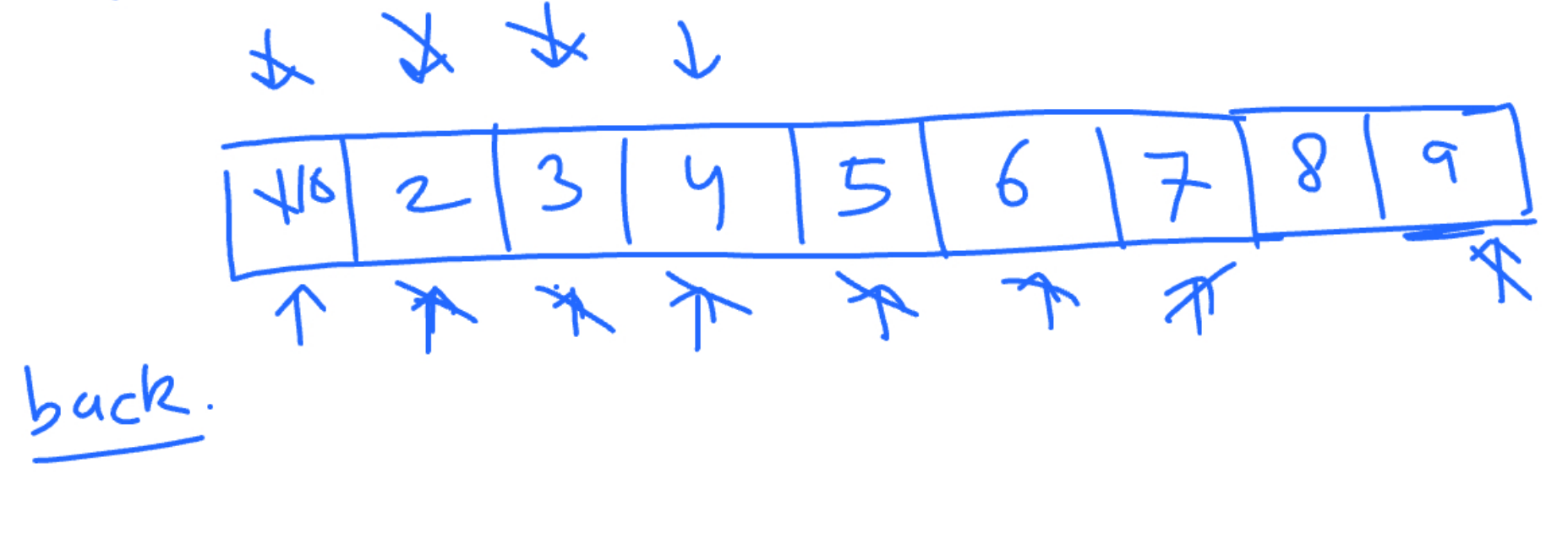
Insert at End (with tail) → O(1)

Delete at Begin → O(1)

Head → O(1).



Push(1)
push(2)
- front() — 1
- pop()
pop()
push(3).
front() —— 3



Push(1)
push(2)
- front() — 1
- pop()
pop()
push(3).
front() —— 3

N=9

[(back +1) % N]

(8+1) % N = 0

TODO: Implement Queue Using a Static Array.

N=1000

```
class Queue {
    private:
        int arr[N];
        int front;
        int back;
        int size;
    public:
        push()    → O(1)
        pop()     → O(1)
        front()   → O(1).
        isEmpty()
        size()
};
```